

Causation and Logic in Neuroscience

Nihil in terra sine causa fit.

—THE BIBLE

Everything connects to everything else.

—LEONARDO DA VINCI¹

The law of causation . . . is a relic of a bygone age, surviving, like the monarchy, only because it is erroneously supposed to do no harm.

—BERTRAND RUSSELL (1992)

After coming to America as a postdoctoral fellow, I gradually learned how strongly our upbringing affects our interpretation of the world. Different cultures shape the brain differently, making it difficult to empathize with a new society that holds viewpoints that are distinct from the social milieu where you spent your formative years. Shaking hands while looking at another's eyes induces different emotions than bowing and expressing submission. Most people agree that our upbringing is linked to our prejudices. I often wondered whether cultural differences can appear not only in moral and emotional issues but also in scientific thinking. Even practices that appear trivial, such as reading and writing from left to right or vice versa, may influence the workings of the brain.²

1. Brulent and Wamsley (2008).

2. Scholz et al. (2009). Learning a second language can induce structural changes in white matter (Schlegel et al., 2012). The brains of musicians are favorite targets of researchers examining practice-induced structural changes in the brain. Use-dependent brain plasticity has been consistently demonstrated in the brains of professional musicians and in children receiving instrumental musical training for several months (Hyde et al., 2009; Lutz, 2009).

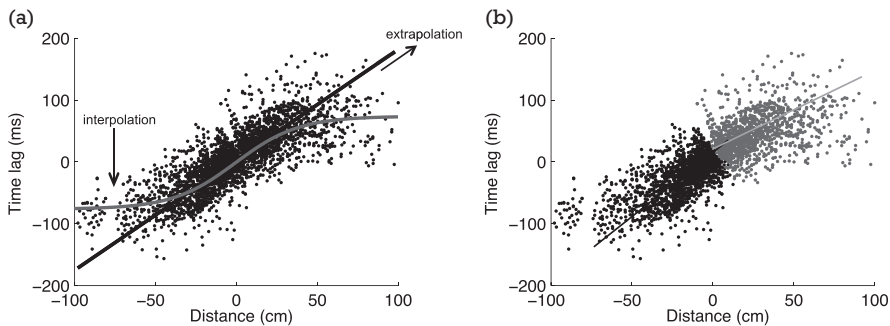


Figure 2.1. Interpreting graphs. A: Each dot corresponds to the measured distance the animal walked in an experimental apparatus plotted against time difference between the spikes of a neuron pair. The black line is a linear regression of all values. Taking into consideration the tails of the distribution, the sigmoid (*gray line*) is a better fit. B: An alternative interpretation is that there are actually two clouds (*black and gray*), which have different relationships between distance representation and the time lag of spikes.

Can we somehow put ourselves outside our subjective world and use objective methods to make valid judgments without being influenced by our pasts? If my reasoning has some merit, then examining the same problems from multiple angles may be useful in understanding the brain. Let's start by evaluating our everyday tools for reasoning in the Western world: logic and correlations and the necessary and sufficient conditions of causation.

OBSERVATIONS, CO-OCCURRENCES, AND CORRELATIONS

Hypotheses in the neurosciences are most often tested by models of associations between observed properties of the world. Figure 2.1 shows a generic example of a relationship between two observed variables: in this illustration, between the distances walked by an animal and the temporal relationships between spikes of neuron pairs corresponding to those distances.³ Inspection of the cloud of dots

3. These data are modified from Diba and Buzsáki (2008), showing the distance measure between place field peaks of a pair of hippocampal place neurons (x coordinate) while the rat ran on a linear track and the time difference between the spikes of the same neuron pair. The physiological explanation of the flat ends of the sigmoid curve is that the distance-to-duration conversion is brought about by theta oscillation dynamics. The duration of the theta cycle is 140 ms (± 70 ms in the figure), which limits the available distance representation to approximately 1 meter in the dorsal hippocampus. To resolve longer distances, the animal must move in the environment (see further discussion in Chapter 7).

suggests a reliable relationship: longer distances correspond to larger intervals between spikes. A simple way to quantify this relationship is to fit a “regression line” over the values (the solid black line). The line is a great simplification, an assumption that all measurements are “represented” by the regression line. The slope of the line and the magnitude of the variance of the values on both sides of the line indicate the reliability and strength of the relationship. This regularity is a “co-relationship” or correlation, in which approximately the same number of values lie above and below the regression line.⁴

While a plot such as that in Figure 2.1 shows only the values of observations, we often wish to read much more from a static cloud of data points. For example, there are many empty areas without measured values on the graph. We can simply assume that those values exist so that, if we had more time and more funding to generate more data, those new points would fill in the blanks between the already-measured ones. This operation, performed intuitively or with mathematical rigor, is called *interpolation* (the arrow in Figure 2.1), but we can also call it *explanation* or *deduction*. *Postdiction* would be another possible term because interpolation is done based on data collected in the past. If we feel comfortable filling the graph with imagined data between the dots, perhaps it is not much of a jump to imagine more data that *extend* the cloud, again following the trend in the observations. This operation is called *extrapolation* or *induction*. Alternatively, we can call it *prediction* or *guessing* from the currently available data, an act of “logical inference.”⁵ It is an abstraction from the present, which amounts to predicting the future. How far can we go in our extrapolation of the graph? Just because two sets of variables correlate now, there is no guarantee that this correlation will continue into the future.

Interpolation and extrapolation, as mathematical abstractions, are independent of the objects or observations being represented. The graph itself is an abstract relationship. The dots have lost their physical correspondences and are now embedded in a mathematical graph. In fact, the same dots could represent

4. The coordinates are both rotationally and translationally symmetric. There is no special need for the y and x axis to point north and east. The graph can be oriented in any direction as long as the relationships between x and y values are unchanged. This symmetry applies to Euclidian space. In Minkowski spacetime, the symmetry applies to time as well (Chapter 10).

5. Conflation among expressions such as “explanation,” “prediction,” “postdiction,” and “cause” is common in interpreting statistical data, but distinguishing them is important for advancing scientific knowledge. The statistical literature often lacks a thorough discussion of the many differences that arise in the process of modeling explanatory versus predictive processes. Explaining (the past) and predicting (the future) are logically different (Shmueli, 2010). To “predict” (from the Latin *pre-* in front of, and *dicere* -to speak) means to verbally declare what is going to happen in the future. Postdiction is an inference. Thus, statistics is not free of subjective interpretation.

any relationship and are thus substrate-free. For example, they could represent an observed relationship between emotional magnitude and neuronal firing. In contrast, the interpretation of the relationship always depends on their substrate. Real things have a physical substance. Because the substrates of the variables are missing from the graph, the experimenter adds them in his or her interpretation to deduce a substrate-dependent mechanism that supposedly explains the observed relationships. This interpretational desire is the reason we give various names, such as *contingency*, *dependence*, *interdependence*, *indication*, *anticipation*, *prediction*, or *forecasting* (I might add *fortune-telling* or *crystal gazing*), to such simple x - y relationships.⁶

In our example in Figure 2.1, we took distance as the independent variable in the tradition of the outside-in framework. The distance between the two place fields in the maze can be measured objectively by a ruler, and the goal is to learn how distance is “represented” in the brain (e.g., by time offsets). In a sense, we are asking how distance predicts duration. However, as you will see in Chapter 7, the brain internally generates time offsets between neuronal spikes, and, with the help of the locomotion speed of the animal, it calculates the distances. In the inside-out framework, the cause is a brain mechanism and the consequence is distance estimation. Distances are meaningless to a naïve brain without calibrating the visual cues with action (Chapter 3). Thus, depending on the framework, the cause-and-effect relationships can change.

However, another experimenter may disagree with our interpretation of the x - y relationship. At the ends of the clouds, there are fewer data points, but those seem to lose the linear relationship characteristic of the other data points. Perhaps a sigmoid function (S-shape curve, in gray) is a better fit. To a third observer, the cloud of dots may appear to be split (Figure 2.1B). When all the dots are lumped together in our mind, the single-cloud interpretation justifies the use of the term “integration” or “pattern completion,” by analogy to completing a picture or a sentence with missing details. But the more experienced eye of the third observer suggests that there are actually two clouds, and, therefore, two relationships exist, a case for segregation or pattern separation. Do the points in the graph reflect a single integrated relationship or two segregated ones? The regression problem then becomes a classification problem. Our scientific vocabulary and views are often formed by data plots even if they are more sophisticated than those shown in this simple figure. Without an arbitrator or supervisor, we cannot tell who is right and who is wrong.

6. Francis Bacon declared that experiments and observations are sufficient to obtain reliable ideas about the world. In contrast, René Descartes, although he had faith in mathematics, argued that bare facts are never self-explanatory: they require interpreting. A graph will never tell how thing happened, only how they *could* have happened.

CAUSALITY AND SELF-CAUSATION IN THE BRAIN

In Figure 2.1, we identified some regularity between two variables: one is the thing-to-be-explained (*explanandum*), called the dependent (y) variable, and the other is the thing-that-explains (*explanans*), called the independent (x) variable.⁷ In more general terms, we can also call them the “defined concept” and the “defining concept.” Of course, our strongest desire is to conclude that one variable predicts or causes the other. Ideally, the defining concept is more general than the defined concept, with some asymmetry between them. “The brain is a machine”; however, the other way around—“the machine is a brain”—does not apply. Accordingly, our choice to call one variable independent and the other dependent is often arbitrary and may reflect our preconceived bias. In some cases, we are able to manipulate the independent variable experimentally and observe its effect on the dependent variable. In neuroscience we often observe quantities which we cannot manipulate directly or for which there is no way to “independently” manipulate a single variable without indirectly manipulating many more properties of the system. In this case, the choice of calling one variable independent and the other dependent is often arbitrary and reflects a bias. In fact, the plot itself is symmetric and its statistical description would be the same if we reversed the x and y coordinates: the ratios would not change. Yet, in neuroscience, we place behavioral observations or stimulus features on the x axis (in this instance, the distance walked) and brain measures on the y axis. This tradition takes its origin from the outside-in framework as discussed in Chapter 1. Putting aside this historical issue, may we be able to make a causal inference from such data by plotting and evaluating the correlational measurements in some other way?⁸ Equally important, we should ask about our confidence in the prediction: the probability of being right or wrong. Are these observational data enough, or do we need to do additional experiments?⁹ To show convincingly that one variable causes the other, we must

7. Hempel and Oppenheim (1948).

8. Our expectations often influence our interpretation. From the outside-in framework, one can assume that the brain “measures” distances in the environment and represents them by temporal offsets between the representing neurons. However, experiments designed to challenge such correlations showed that the temporal offsets between hippocampal neurons are determined by internal rules of brain dynamics that remain the same across increasing size environments. As a result, the spatial resolution of the neurons decreases in larger environments. Distance representations in the hippocampus scale with the size of the surrounding world (Diba and Buzsáki, 2008). Thus, brain network dynamics is the independent variable that explains distance, not the other way around.

9. What if we remove Jupiter from the solar system? One astronomer would say that we can calculate the consequences from the available data (no experiment is needed). But another

vary the independent values experimentally and examine how the dependent variable follows such changes.¹⁰ Direct manipulation of physical objects is the most basic way for arguing the persuasive rule of cause and effect. When a child discovers that flipping a light switch can make the room appear and disappear, he repeatedly practices it to learn that he is the sole causative agent of the consequences. Similarly, when a rat presses a lever that delivers food, it repeatedly tests the consequences of its action. It seems that functioning in the world requires animals to develop the concept of cause–effect relationships.

Both mathematics and scientific thinking require logic. All mathematical theories can be reduced to some collection of axioms or set of rules on which an internally consistent system is built. These axioms should not be violated. But science is supposedly different. It has no *a priori* rules. We simply observe regularities in nature and call them “laws.”¹¹ When we interfere with those regularities, we become part of the relationship and therefore may alter it. As Heisenberg, the father of uncertainty in quantum mechanics, famously stated: “What we observe is not nature, but nature exposed to our method of questioning.” This healthy skepticism is important because our measuring instruments may affect the cause–effect relationship.

Every action we take has an effect on the system we wish to observe. However, this cause-and-effect situation runs deeper than just a difficulty in experimental science—it forms the root of scientific inference in the Western world: once the cause is revealed, the problem is “explained,” the truth is found. In the traditional wisdom, there are no effects without causes, and no logical argument can be delivered without causation. Western philosophy is built on a foundation of causality. Yet much of the world has moved forward and onward

astronomer would disagree and claim that the impact is unpredictable given the complex relationships among the celestial bodies (and many more such perturbations, such as removal of other planets or perhaps galaxies, are needed to get a firm answer). These are the types of questions that neuroscience constantly faces.

10. Without manipulating the system experimentally, one can add or leave out variables to “re-gress out” noncausal agents. It is never foolproof, but often this is the best one can do without interfering with the system. Descriptive statistics that reveal correlations are a bit like passive associations. In neither case can one make causal grounding. The source of knowledge is in the manipulation.

11. Laws, as organization tools of scientific thinking, have specific theological associations. Physicists, including Newton, who introduced these laws have assumed that things are “ruled” or “governed” by God, who was regarded as changeless and eternal. In Christian and many other religions everything must have a cause and laws and Order of Nature are assumed to be fixed in their forms (Toulmin and Goodfield, 1965). However, laws are substrate-free. They are not physical things, so they cannot exert an effect on anything else, short of divine interventions. They simply reflect regularities—the earth orbits around the sun.

without accepting such causality rules. The idea of causation is quite different in Asian cultures than in European cultures. For example, in Buddhist philosophy, nothing exists as a singular and independent entity because things and events depend on multiple, co-arising conditions.¹² A Buddhist monk or teacher lights three incense sticks leaning against each other, as a demonstration that if any of the sticks falls, the others will fall as well. Similarly, Confucians approached causation in an all-encompassing manner, searching for a network of connections. Historical changes are explained by coincidences of events instead of a particular cause. In Muslim philosophy, causal relations have no inherent necessity at all. Instead, they are just expressions of what God does.¹³ Therefore, it should not surprise us that people from different cultures—yet with the same kind of brains—judge the affairs of the world surrounding us differently.¹⁴ So why do we assume that the European philosophy of science is the superior or only way to view the world and ourselves, especially given that many other cultures have achieved great scientific discoveries? Is it really true that when the cause is identified, the truth is found? Even Descartes hesitated to accept such a link. First, he states: “If a cause allows all the phenomena to be clearly deduced from it, then it is virtually impossible that it should not be true,” but then he immediately backtracks and adds: “Nevertheless, I want causes that I shall set out here to be regarded simply as hypotheses” because we often make “assumptions which are agreed to be false.”¹⁵ The modern meaning of the term “hypothesis” refers to a conjecture which can be rejected by observations and experiments,

12. “Origination” (or *pratityasamutpada* in Sanskrit) refers to the principle of interdependent causation and explains the twelvefold chain that describes the endless rebirth of life (Dalai Lama and Chodron, 2017). In metaphysics, *accidentalism* denies the doctrine that everything occurs or results from a definite cause.

13. Leaman (1985).

14. Carl Jung also rebelled against the Kantian axiom that “nothing can happen without being caused.” Instead, he argued for “an acausal connecting principle,” or the principle of “meaningful coincidences,” based on Eastern philosophies. He and the physicist Wolfgang Pauli coined the term “synchronicity” to replace causality as a principle of explanation. Explanation “cannot be a question of cause and effect, but of a falling together in time, a kind of simultaneity” (Jung, 1973). Jung explained everything with synchronicity, including telepathy. For a statistician, coincidence is not a surprising, meaningful event because “with a large enough sample, any outrageous thing is likely to happen with no apparent causal connection” (Diaconis and Mostelle, 1989). These issues are eloquently exposed in Arthur Koestler’s book *The Roots of Coincidence* (1973). [NeuroNote: Koestler attempted suicide in Lisbon in 1940 after he failed to return to England, but he survived. After he developed Parkinson’s disease in his seventies, he and his wife committed double-suicide.]

15. Descartes (1984).

thus eliminating potential causes rather than proving the truth. Scientific knowledge is tentative and not definitive; it is progressive but not absolute.

Cause and Effect

Causality is a subtle metaphysical notion, an abstraction that indicates how events in the world progress. The common notion of causation can be traced back to Aristotle and is a statement about how objects influence each other across space and time. From the correlation graph (Figure 2.1), we cannot make the claim that x is the cause of y without extra manipulations.¹⁶ Causation is defined as a chain of dependency, where the cause always precedes the effect: the cause is in the past of a present event. An event, A , is a cause of event B if the presence of A is necessary for B to happen. For A to have any effect on B , event A has to happen first.¹⁷ David Hume remarked that “if the first object had not been, the second never had existed.” In a deterministic framework, the presence of the cause is a necessary and sufficient condition for an effect to happen. Causality and determinism are often intertwined. Of course, an effect can have many co-occurring causes, but we are compelled to believe that we should be able to identify at least one. For example, when an individual gene alone cannot account for a mental disease, it can still be viewed as a conditional cause when co-occurring with other causes. When enough necessary components occur together, they may be sufficient to induce the disease. Under this framework, the triad of correlations and necessary and sufficient conditions complement each other. In another formulation, one can distinguish an *immediate* or *proximal cause* (also called *precipitating cause*: a stroke in the motor area caused paralysis), *predisposing cause* (she got upset), and *sustaining cause* (she had high blood pressure). Neither the predisposing nor the sustaining course alone could be designated as necessary and sufficient condition. Yet the proximal cause may not have happened without the sustaining and predisposing causes.

16. One step forward in interpreting a graph is to vary the independent variable. In this case, the designation of the independent and dependent variables is critical because regression of x on y is not the same as regression of y on x .

17. This logic stems from a Newtonian framework, which has a “time arrow.” When this fundamental requirement of causation (i.e., the notion of time) is removed from its definition, things become rather complicated. In quantum-mechanical examples—such as entanglement—two particles can have interdependent or “entangled” properties. The usefulness of “classical” causation and the role of time as part of causality are intensely debated in both general relativity and quantum mechanics (Hardy, 2007; Brukner, 2014; Ball, 2017). As logic is timeless, it is not clear why time should be part of causation. We visit the nature of time and its alleged role in brain operations in Chapter 10.

Although Aristotle's formulation of the cause-and-effect relationship has undergone extensive revisions over the past millennia, it has remained the main tool of reasoning in the laboratory, likely due to its simplicity. However, even in everyday life, one can often see how such logic fails. My shadow follows me, and without me it never occurs. My shadow and I (often) correlate. I am the necessary (but not sufficient) condition because my shadow is not always detectable. In darkness, the suspected causal rule is violated because the occurrence of my shadow is conditional upon the presence of light. However, light is not a cause and not even a co-contributing component but a condition.¹⁸

Similar violations are found in neuroscience. A weak stimulus, such as an ant climbing on my leg, can sometimes induce reliable firing of neurons in multiple brain circuits and is prominently noticed (an effect). At other times, the same cause (the ant) may not induce a noticeable effect at all (for example, when I am intensely preoccupied with writing this book chapter). Again, the change in brain state is not considered a co-cause, but it is a necessary condition for the detectable cause–effect relationship. Later, I highlight some of the pitfalls of our everyday reliance on causation as a scientific rule. My goal, of course, is not to disqualify formalized reasoning but to call attention to why the simple recipe of logic fails in neuroscience, especially when things become really interesting.

Because of the omnipresence of logical inference and causation in science, both philosophers and scientists have undertaken multiple rounds of explanations and attempts to refine them. Each time, these concepts were approached with a particular point of view and purpose, and, not unexpectedly, the resulting explanations vary. Philosophers are primarily interested in the ontological question of whether causation is an inherent property of the physical world independent of the human mind or whether it has epistemic limits. As we learned in Chapter 1, David Hume's subjectivism separated the objective world from our first-person, subjective experiences and introduced the idea that our mind transforms and “represents” the objective reality. He criticized rationalists who assumed that we can acquire knowledge of the world by trusting our minds. For instance, we observe a regular relationship between lightning and thunder, but the inference that lightning causes thunder is only a subjective mental operation, so there may be no objective causality between

18. In Boolean logic, it could be called a “gate.” In algebra, Boolean logic follows Aristotelian logic in a formalized manner, where all values are reduced to a binary true (1) or false (0). Everything in the middle is excluded or forced to belong to the true or false category (you either perceive something or not). The binary nature of Boolean algebra is what makes it so suitable for computer science, using the true and false statements with the fundamental operators And, Or, Xor and Not. Boole's symbolic system makes computation and logic inseparable. Early brain models assumed that brain operations follow a Boolean logic (McCulloch and Pitts, 1943).

the two. Hume thought that we do not experience the cause itself; we simply infer it from the regular appearance of the cause and effect. Immanuel Kant countered Hume's interpretation by arguing that there is a universal principle of causality that cannot be based on experience. Kant promoted the idea that the realm of things in themselves (*das Ding an sich*) is the cause of the realm of appearances (phenomena).¹⁹ Our experience is shaped by a priori causality because the mind interprets the sensory inputs arriving from an objective, observer-independent world.

Hume's idea of representation has shaped the thinking of cognitive psychologists and neuroscientists alike and became the de facto scientific explanation of how "attributes" of objects become neuronal "representations." His subjectivism has been modernized to a subjective realism in which the mind represents, but does not faithfully mirror, the outside reality. Therefore, my representation of truths and judgments can only become similar to your representations through a mutually agreed consensus. Each brain in isolation would generate a somewhat different representation of the same object, color, or sound. If objective and subjective reality are separate, then our mental influence of causation does not necessarily apply to the relationships present in the physical world.

In a deterministic system with one-way interactions among the constituents, cause-and-effect relationships can be reliably described by Aristotelian–Humean arguments because the cause regularly precedes and induces the effect.²⁰ Determinism heavily rests on causality. In the Newtonian world, cause-and-effect relationships seem to work well since nothing moves unless it is being moved.²¹ If the initial conditions and all variations are known in a system, in principle, all consequences can be accounted for. However, in practice, our

19. Interestingly, Muslim theology makes a similar distinction between *batin* (things-in-themselves or *noumena*) and *zahir* (phenomena). Muslim scholars believe that categories of causality and existence apply only to the noumenal realm, but not to *zahir*. Muhammad exists, but we do not know how (El-Bizri, 2000); we just have to believe it. For a concise exposure to fresh thinking about causation, see Schaffer (2016).

20. The concept of cause, as usually presupposed in standard causal modeling, is a "mover" and cannot be used to reduce probability. Can the absence or reduction of something be a cause? Inhibition in the brain is such a case. The primary role of a special, dedicated group of neurons in the brain is to inhibit excitatory neurons and each other. Thus, elevation of firing rates of excitatory cells can be brought about by excitation via other excitatory neurons or by reduction of inhibition ("disinhibition cause?").

21. Nearly every tool and machine is based on the logic of cause-and-effect relationship. Machines may represent the exteriorized reflection of this logic (Chapter 9), or, conversely, the functioning of the tool and machine components could have led to internalization of the observed logic of things (Chapter 5).

knowledge of the real world is limited by inevitably noisy measurements with limited precision, and such wisdom does not offer much solace. Even if we did have complete knowledge of the universe's past, the time needed to compute the future might be as long as the elapsed past.

Troubles with Causation

Problems with causality surfaced early in contemporary physics with the birth of quantum physics, nonlinear dynamics, chaos theory, and quantum entanglement. Quantum physics underwent a multitude of interpretations, including deterministic causality. On the other hand, in the framework of general relativity, time became symmetric and the foundations of cause–effect logic were shaken.

To understand complex economic data, Clive Granger reintroduced the time-asymmetry axiom (Chapter 10) and a novel statistical method for testing causality. “The past and present may cause the future, but the future cannot cause the past.” His method provides predictability, as opposed to correlation, to identify the cause–consequence relationship with the key requirement of separability; that is, that information about a causative factor is independently unique from the variable. Strictly speaking, Granger's method does not present a definition of causality; it only defines a metric by which we can measure how much knowledge of one signal can enable you to predict another signal at a later time. It is ambiguous to the underlying source of predictability.

Yet the absence of the cause and effect between activities in brain regions does not prevent neuroscientists from talking comfortably about causal inferences in correlational data from the brain-derived data all the time. Granger's method has become popular in neuroscience in attempts to interpret cause-and-effect relationships, especially in complex functional magnetic resonance imaging (fMRI), electroencephalograms (EEG), and magnetoencephalographic (MEG) data.²²

22. Statistical models used for testing causality are almost always association-based models applied to observational data. The theory itself is expected to provide the proof of causality, but, in practice, the ground truth is often absent (see Granger, 1969; Wiener, 1956; Mannino and Bressler, 2015). Causality as a tool of inference has grown from a nebulous concept into a mathematical theory with significant applications in many fields of sciences (Pearl, 1995). In addition to Granger's causality measures, other methods, such as directed coherence, partial coherence, transfer entropy, and dynamic causal modeling, have been used to assess the direction of activity flow in the brain (Barnett et al., 2009; Seth 2005). “Proper” statistical methodology for testing causality has been advanced, such as specially designed experiments or association-based, specialized causal inference methods for observational data, including causal diagrams (Pearl, 1995), discovery algorithms (Spirtes et al., 2000), probability trees

But many situations common in neuroscience will result in Granger's formalism giving the incorrect answer. Consider the following real-life observation. In recording from groups of neurons in area *A* and area *B* in the brain, we find a reliable correlation between the peaks of spiking activity of the populations, with neuronal spikes in area *A*, on average, consistently and reliably preceding the peak spiking activity in area *B*. Logic and Granger's formula dictate that it is likely that activity of neurons in area *A* causes the increased activity in area *B*. But we may be surprised to learn that axons of neurons in area *B* innervate neurons in area *A* unidirectionally; in short, area *B* is upstream to area *A*. To explain the contradiction between the anatomical ground truth and the physiological observations, let's assume that neurons in area *A* respond to inputs from *B* with a low threshold (i.e., before neurons in area *B* reach their population peak activity) and stop firing after their first spike. This results in peak spiking of the population in area *A* *earlier* than the peak in area *B*! Neuronal patterns often have such properties. In fact, it is a basic feature of a common feedforward inhibition motif found throughout the brain. Neurons can stop firing under continuous and even increasing drive because intracellular mechanisms may prevent sustained firing in neurons of area *A* (such response is known as *spike frequency adaptation*) or because strong feedback from inhibitory neurons prevents the occurrence of further spiking. Without knowing the anatomical connections and biophysical/circuit properties of neurons, Granger's formula gives the false conclusion that area *A* causes activity in area *B*. This example illustrates just one of the many pitfalls that have to be considered in any "causal" analysis.²³ Of course, this example does not disqualify the existence and importance of cause-and-effect relationships; it only reveals that the method is not always appropriate.

Hidden common causes are the most frequent sources of misinterpretations based on correlational arguments. Take as an example the observation that

(Shafer, 1996), propensity scores (Rosenbaum and Rubin, 1983) and convergent cross mapping (Sugihara et al., 2012). Applying these methods to the brain often requires the assumption that the brain is a deterministic, nonlinear dynamic system, although this assumption is questionable (Friston et al., 2012).

23. Another striking illusion of causality has been described in the entorhinal-hippocampal circuits. While superficial layer neurons of the entorhinal cortex are known to innervate hippocampal neurons but not vice versa, population activity in the downstream hippocampal neurons within theta oscillations cycles peak *earlier* than peak activity in the upstream entorhinal neurons (Mizuseki et al., 2009). The explanation in this case is that it is the fast spiking minority of neurons that fire early in the theta phase, and they have a much stronger impact on their target interneurons than the majority of slow firing cells. We discuss in Chapter 12 that skewed distributions are the main reason that mean population-based calculations are often misleading.

springtime brings flowers and higher birthrates. Nobody argues against this reliable relationship. However, even if this relationship is acknowledged, the issue of causality remains ambiguous because the likely cause is not that spring induces labor, but that couples spend more time together during the long dark nights of winter. Similarly, there is a reliable correlation between the blooming of flowers and birthrate increases. If we are not aware of the nonrecorded “third-party” variable (spring), we may suspect a causal relationship between these two variables as well. This example may seem silly for an easily relatable relationship such as flowers and birth rate, but when we are observing variables for which we do not have the luxury of everyday intuition, the reader can see how easy it would be to mistake correlation for causation. Hidden or permissive common causes are the most frequent sources of misinterpretations based on correlational arguments.²⁴ For these reasons, deterministic causality has been challenged in several other areas of science as well.

Probabilistic Causation

Events can follow each other with high probability, very low probability, or anything in between, so probabilistic influences fluctuate over a wide range from 0, meaning chance, to 1, meaning certainty. However, it is not obvious which aspects of the distribution are “causal” and which are not. Probability is an abstract mathematical concept and not a “mover” needed for causation in Newtonian physics. Determining probabilistic causation is a two-step process: the first step is the description of patterns and probabilistic facts (e.g., the substrate-free illustration of Figure 2.1) and the second is to draw conclusion from the measured probabilities. This second step is not inherent in the observed facts, however, but is an interpretation of the experimenter. When we observe a regular succession between things, we feel compelled to assume a cause. But how likely does succession have to be to reach a threshold to be perceived as “causal”? Might your threshold be different from mine? We single out high- and low-probability events from a continuous distribution and call them cause and chance. Confusion about chance, low probability, and high probability is a major contributor to wishfully interpreted “causal” relationships between gender and performance, genes and intelligence quotient (IQ), and religion and moral values. Conflation between explanation (postdiction) and

24. Calcott (2017) discusses the distinction between permissive and instructive factors. These factors are not really causal, yet their presence may often be critical. *Permissiveness* can be viewed as a gating function that allows the cause to exert its effect. Instructive factors may be a prerequisite but can induce a final effect only in conjunction with other factors.

prediction is also common in neuroscience data. Yet correlating, explaining, predicting, and causing are different.

Like causality, the notion of probability is often entangled in metaphysical disputes, and its interpretation appeals to mysterious properties.²⁵ Such considerations led Bertrand Russell to suggest that we remove causation from scientific thinking: “In the motions of mutually gravitating bodies, there is nothing that can be called a cause, and nothing that can be called an effect; there is merely a formula.”²⁶ In the mathematical equations of physics, there is no room or need for causes. Do we need them for neuroscience? Paradoxically, Russell’s quantitative probabilistic formulation managed to save rather than eliminate causation as the tool of scientific explanation. The concept of probabilistic causation replaced deterministic substrate-dependent causation.²⁷ Spurious regularities, a headache for deterministic causality, had become quantifiable and therefore could be contrasted with “true” regularities. However, making causal inferences from correlational data is harder than is usually thought, mainly because probabilistic causation is a statistical inference, whereas causation should involve physical substrates (“enhanced neuronal synchrony is the cause of epilepsy”). The correlation between various genes and schizophrenia leaves no doubt that aberrations in such genes increase the probability of the disease. On the other hand, claiming that my friend’s schizophrenia was caused by particular genes is hard to support. Correlations need to be “challenged” to improve their explanatory power. Adding true meaning to correlations requires a second validation step, a hypothesis-testing practice based on some perturbation of the explanatory independent variable. For this reason, we need to do many independent experiments and worry more about replicability than causal explanation.

25. Despite the solid mathematical bases of probability and statistical inference, the connection between probability relations and correlational dependencies is often vague.

26. Russell et al. (1992). The cause (C) has a causal influence on the effect (E) if conditional probability ($E \mid C$) for E reliably changes under free variation of C . In an open, linear system, this requirement is not an issue, but in complex interconnected systems, such as the brain, the assumption of free variation of C is most often not tenable.

27. Moyal (1949); Churchland (2002); Mannino and Bressler (2015). While the term “probabilistic” is a useful theoretical construct, its use in real life may not be always practical. Consider Richard Feynman’s explanation of the increase of entropy as defined by the second law of thermodynamics, which says that entropy is always increasing. “Things are irreversible only in a sense that going one way is likely, but going the other way, although it is possible and is according to the laws of physics, would not happen in a million years” (Feynman, 1965, p. 112). Similarly, in probabilistic causation, everything can be explained by singular causes in principle, but most of them may be statistically and practically improbable.

MUTUAL AND SELF-CAUSATION

Probabilistic causation appears more attractive and perhaps more appropriate for describing interactions in complex systems such as the brain. After all, it is only in interactions that nature affects the world. One may wonder whether transforming the philosophical problem to a precise mathematical description makes the word “causation” more valid. This issue is especially relevant in mutually interconnected and strongly recurrent complex systems with amplifying-damping feedback loops and emerging properties. Brain networks are robust and balanced, so sometimes even very strong stimuli may fail to affect the brain. Other times, minute perturbations can have a large impact on neuronal activity depending on the state of these networks. The most important feature of complex systems is perpetual activity supported by self-organized dynamics (Chapter 5). Ever since an EEG signal was first detected from the surface of the brain and scalp, its ever-changing electric landscape has been called “spontaneous.” Related terms, such as *endogenous*, *autogenous*, *autochthonous*, *autopoietic*, *autocatakinetic*, or *self-assembled* are more popular in other complex systems, but they refer to the same idea.²⁸ These concepts have proved to be difficult to tackle because, by definition, the generation of spontaneous activity occurs independently of outside influences. Thus it must be induced by some self-cause, not by external causes, although the latter are much easier to study.

How can we separate causes from effects in self-organized brain networks with multiple parallel and interacting feedback loops? In many cases, brain patterns, such as neuronal oscillations, are not caused but are released or permitted. For example, sleep spindles in the thalamocortical system can be induced occasionally by some sensory stimuli but most often they just occur because the decreased activity of subcortical neuromodulators during sleep do not prevent their occurrence. Spontaneous patterns emerge and evolve without obvious trigger events. A pertinent analogy is the emergence of a tune from the interaction of jazz musicians. When they find a coherent melody, it decreases the degree of freedom of each player and forces them to time their actions to the emergent tune. But—and this is where it gets interesting—emergence is not the sum of multiple decomposable physical causes but a result of a multitude of *interactions*. Emergence is not a “thing” that affects neurons or humans.²⁹

28. The term “self-organization” was introduced by the British psychiatrist W. Ross Ashby (Ashby, 1947). Increasing order in self-organized systems is discussed in Kampis (1991), an accessible text on self-organization. For a more complex treatment of the subject, see Maturana and Varela (1980).

29. Hermann Haken (1984) introduced the term “synergetics” to explain emergence and self-organized patterns in open systems far from thermodynamic equilibrium. His “order

Just because two sets of data move together over time, that does not necessarily mean they are connected by a causal link. Correlations are not causation, as conventional wisdom rightly says. Therefore, cause does not reduce to correlation. To understand a machine, it is not enough to look at it carefully. We must take it apart and put it together or remove parts from it and see what happens. Of course, in a closed system, such as a car, if all parts and all their linear interactions are known, in principle, the state of the system can be described at any time. However, in open systems, such as the brain with its complex dynamics, such complete description is rarely possible, mainly because the interactions are non-linear and hard to predict.

Challenging Correlations

Perturbation, a targeted challenge to the system, is a powerful method for investigating a system provided that perturbation is done using a correct design and the resultant changes are interpreted properly.³⁰ In neurology and brain sciences, the perturbation method is one of the oldest. Damage or degeneration

parameter” is the synergy or simultaneous action of emergence. In synergetic systems, upward (local-to-global) and downward (global-to-local) causations are simultaneous. Thus, the “cause” is not one or the other, but it is the configuration of relations. One can call such causation reciprocal or circular, terms that are forbidden by Aristotelian and Humean logic (*circulus vitiosus*). However, in complex systems, circular causation is neither vicious nor virtuous but a norm. Reciprocal interactions weave the correlations. An excellent discussion of synergetics in neuroscience is by Kelso (1995), with an abbreviated version in Bressler and Kelso (2001).

30. Controlling neuronal activity in real time was only a dream for neuroscientists not so long ago. Since the introduction of optogenetics into mainstream neuroscience by Karl Deisseroth and Ed Boyden (Boyden et al., 2005), we now have the ability to turn on or off genetically identified neurons at will. This revolutionary tool allows us to challenge long-held views and generate new advances in our understanding of brain function in both health and disease. Unfortunately, when new techniques are invented, optimism and hype often trump humility. Optogenetic techniques are too often advertised as causal tools to discover the relationship between activity patterns in specific neuronal circuits and behavior. However, because neurons are both embedded in circuits and contribute to circuit function, their perturbation can bring about secondary and higher order changes, which need to be separated from the primary action of optical stimulation (Miesenbock, 2009). This separation is rarely straightforward because the network may change in an unpredictable manner. For this reason, it is not easy to tease out whether the observed variable is caused by silencing of a specific group of neurons or is the unseen consequences of novel patterns in their targets. Without knowledge of the precise effect, the assumed cause ceases to be a true cause. A potential improvement is the combination of correlation and perturbation methods. To fully exploit the advantages of a perturbation technique for brain circuit analysis, one should try to limit the perturbation to a small number of neurons that are continuously monitored together with their neighbors and upstream targets (Buzsáki et al., 2015; Wolff and Ölveczky, 2018).

of certain brain parts as a result of injury or disease has offered numerous and powerful insights into the contribution of various brain structures to normal function, suggesting that different parts of the brain are specialized for different functions. Complex functions, such as memory, emotions, and planning, have found their homes in the neural substrates of the hippocampus, amygdala, and prefrontal cortex, respectively, with the often explicit conclusion that neuronal activity in these structures is causally related to the respective behaviors. In animal experiments, structures or their parts can be deliberately damaged, “turned on,” or “turned off” as a “causal” manipulation. The ensuing behavioral, cognitive deficit can then be interpreted to indicate that, in the intact brain, the perturbed structure is responsible for that behavior.

But slam on the brakes. Such a conclusion is not always warranted. For example, massive degeneration of substantia nigra neurons (loss of function of dopamine-producing neurons) is followed by muscular rigidity, slowed movement (bradykinesia), and tremor. However, decreased dopamine is a necessary but not always sufficient condition for the symptoms to occur, and similar symptoms may also occur in the absence of dopamine dysfunction. Similarly, subcortical denervation of the hippocampus in rodents results in more severe cognitive deficits than damage to the hippocampus itself. This happens not only because the hippocampal circuits can no longer perform their physiological jobs but also because the denervated hippocampus induces abnormal neuronal activity, adding insult to injury. Conversely, if behavior is unaffected long after lesioning a particular brain structure or circuit, one cannot rule out its important contribution to that behavior because the brain circuits have redundancies, degeneracies, and plasticity that can often compensate for an acute impairment. These examples illustrate why, in a densely interconnected dynamical system like the brain, even minor and local perturbations may induce unexpected activity in downstream and distant structures. Furthermore, functional recovery may occur via alternative behavioral strategies and adaptive repurposing of structures that are targets of the damaged area. The emerging nonphysiological activity in these downstream structures can be as damaging as the absence of their upstream partners. These considerations point out the caveats and limitations of using causation in our search for physiological function in the complex labyrinth of brain dynamics.

In conclusion, statistical correlations are similar to associations. In neither case can one rely on causal grounding, which requires active manipulations. In the strongly interconnected complex networks of the brain, perturbation may bring about secondary and higher order changes, which need to be separated from the primary effects. This is rarely straightforward as complex networks often respond to challenges in an unpredictable manner. Since there is no simple recipe for how to deal with emergent systems, correlational and

perturbation methods should be combined whenever possible because they represent complementary ways of analyzing the same phenomena. This is the best we can do.

The difficulty in applying causation as the sole tool for inference will often reappear in this book. In the next chapter, we will examine the relationship between perception and action. The two are often correlated. As we learned in this chapter, correlation is symmetric—in the mathematical sense. However, in searching for a cause with an assumed direction, we designate one set of variables independent and the other dependent and make the relationship asymmetric in our minds. To verify our intuition, we must vary one of the variables and quantify the consequence on the other variable. From this perspective, there should be a distinction in whether we talk about a perception–action cycle (where perception is the critical variable for action) or an action–perceptions cycle (where action is critical for perception). While in everyday parlance such distinction may sound prosaic and hair-splitting, I submit that the failure to recognize the importance of this difference is a main contributor to why the outside-in framework continues to dominate cognitive neuroscience. In an uncalibrated (naïve) brain, changing the sensory inputs may not impact the motor response much (save phylogenetically learned fixed action patterns), whereas moving the sensors by actions brings about large changes in the activity of sensory structures. The next chapter discusses how confusion between cause and effect has impacted thinking in perception research.

SUMMARY

In neuroscience, we often compare our observational data against association-based models, such as correlations, and plot our data as a relationship between presumed independent and dependent variables. When we identify some regularity between two variables (find a correlation), we are tempted to regard one set of variables as the thing-to-be-explained and the other set as the thing-that-explains. This is the basic logic of cause-and-effect relationships. Interpreting correlations is a two-step process. The first step is the description of facts (the correlation), followed by a second, biased step—an assumption of a cause-and-effect relationship from a symmetric statistical relationship.

Causality is the most critical pillar of scientific inference in the Western world. Revealing a cause amounts to an explanation. However, other cultures

that do not rely on cause-and-effect arguments can also arrive at valid scientific conclusions. The concept of causation is especially problematic in self-organized systems with amplifying-damping feedback loops, such as the brain. Causes in such systems are often circular or multidirectional; events are not caused but emerge from the *interaction* of multiple elements.

